

ORNSKOLDSVIK, Sweden

WMO: 022670

Lat: 63.412N

Lon: 18.980E

Elev: 354

StdP: 14.51

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.0	-0.7	-11.0	3.1	-5.8	-5.5	4.1	-0.3	26.2	35.2	22.8	35.3	4.1	310	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	77.1	62.2	73.6	60.5	70.4	59.2	65.0	73.5	62.7	70.5	60.8	67.6	9.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	84.1	67.5	59.5	77.0	64.9	57.6	71.8	63.4	30.1	73.2	28.4	70.4	27.1	67.6	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.7	16.3	-13.6	81.9	5.4	4.9	-17.5	85.4	-20.6	88.2	-23.6	91.0	-27.5	94.5		
(5)			-14.0	67.3	5.3	2.1	-17.8	68.8	-21.0	70.0	-23.9	71.2	-27.8	72.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	20.7	20.3	26.9	36.5	46.1	54.2	60.3	57.2	49.5	38.8	30.9	25.6		
	DBStd	15.53	9.62	11.09	8.55	5.78	7.09	5.33	5.10	4.98	5.24	5.98	7.43	8.87		
	HDD50	4804	909	833	716	405	166	20	0	6	72	348	573	756		
	HDD65	9513	1374	1253	1181	855	587	327	166	248	466	812	1023	1221		
	CDD50	793	0	0	0	0	44	145	319	228	56	1	0	0		
	CDD65	26	0	0	0	0	0	1	20	5	0	0	0	0		
	CDH74	280	0	0	0	0	8	39	203	30	0	0	0	0		
	CDH80	41	0	0	0	0	0	7	33	1	0	0	0	0		
	WSAvg	7.2	7.5	7.1	7.7	7.0	7.3	7.8	6.8	6.5	7.0	7.2	7.2	7.6		
	PrecAvg	26.1	2.0	1.3	1.3	1.3	1.5	2.0	3.4	3.4	2.6	2.6	2.5	2.0		
(15) Precipitation	PrecMax	41.1	4.2	3.0	3.4	3.5	2.5	4.5	7.7	6.1	6.1	7.2	7.2	4.4		
	PrecMin	17.6	0.3	0.2	0.3	0.2	0.1	0.7	0.3	0.6	0.7	0.7	0.7	0.2		
	PrecStd	5.0	0.9	0.7	0.7	0.8	0.7	0.8	1.8	1.7	1.5	1.5	1.7	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	43.2	49.7	63.2	74.6	78.8	83.9	77.3	67.9	56.1	48.7	43.2		
		MCWB	37.7	38.1	41.3	49.4	56.9	60.4	65.0	64.8	58.1	49.2	45.7	40.2		
	2%	DB	37.6	41.0	44.8	57.3	69.9	72.8	79.3	73.6	64.2	52.8	44.9	40.8		
		MCWB	35.4	37.1	38.1	46.0	55.8	57.3	63.5	61.8	55.8	48.6	43.5	38.0		
	5%	DB	35.8	37.4	41.8	51.7	64.8	69.0	75.5	70.4	61.6	50.2	42.8	38.6		
		MCWB	34.3	34.7	36.2	42.0	52.8	56.4	62.2	60.1	54.7	47.1	41.3	37.0		
	10%	DB	33.9	34.0	38.9	47.8	60.3	66.0	72.0	67.6	59.2	48.4	41.0	37.0		
		MCWB	32.8	32.6	34.5	39.9	49.4	54.5	60.7	59.1	54.0	45.8	39.5	35.6		
	0.4%	WB	37.9	38.8	41.7	50.2	59.4	62.4	68.6	68.0	60.1	51.4	46.5	40.5		
		MCDB	39.5	42.0	48.0	61.3	70.5	72.9	74.7	72.5	64.3	53.7	47.7	41.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	37.1	38.8	46.3	56.5	60.1	66.0	64.2	58.0	49.5	43.6	38.4		
		MCDB	36.8	40.5	43.8	56.2	67.9	69.6	75.0	70.5	61.9	51.5	44.6	39.4		
	5%	WB	34.4	34.8	36.7	42.8	53.3	58.0	63.9	61.8	56.4	47.7	41.6	36.8		
		MCDB	35.4	36.6	40.8	50.4	63.7	66.4	73.0	67.4	59.9	49.9	42.5	37.9		
	10%	WB	32.7	32.7	34.8	40.3	50.4	55.9	62.0	60.1	55.0	45.8	39.7	35.3		
		MCDB	33.6	33.6	38.2	46.7	58.8	62.9	70.1	65.2	58.2	47.8	40.6	36.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	13.1	15.0	16.6	18.3	17.8	17.3	16.6	15.4	11.6	9.2	10.8		
		MCDBR	11.7	12.4	17.1	25.1	26.1	24.0	23.4	21.9	19.5	14.8	9.5	11.3		
		MCWBR	11.0	10.2	12.1	15.3	14.3	12.3	11.4	12.0	12.6	11.4	8.7	10.5		
	5% WB	MCDBR	11.1	10.4	15.8	23.4	24.3	20.4	20.1	18.0	16.1	13.0	9.2	10.6		
		MCWBR	10.5	8.7	11.7	15.1	14.1	11.7	11.3	10.7	11.6	10.6	8.8	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.283	0.307	0.345	0.346	0.344	0.361	0.351	0.320	0.305	0.367	0.348		
	taud		2.048	2.202	2.282	2.348	2.457	2.495	2.466	2.488	2.558	2.432	2.721	2.647		
	Ebn at Noon		125	205	250	265	276	279	269	262	250	201	117	70		
	Edh at Noon		9	19	26	31	30	30	30	27	21	16	8	5		
(44) All-Sky Solar Radiation	RadAvg		49	236	706	1189	1572	1787	1642	1245	757	308	79	24		
	RadStd		6	44	78	110	173	117	163	106	83	55	17	2		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-229	N/A	N/A	N/A

Nomenclature: See separate page